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1  cd "F:\ECON 408\Practice do file"
2
3
4  * (1) Biased ARMA(2,1) Model Simulation
5  *-----
6
7  clear all
8  set seed 1000
9
10 capture postclose buffer
11 postfile buffer beta_L1 beta_L2 alpha_L1 using arma22_misspecified.dta, replace
12
13 forvalues i = 1/1000 {
14     qui drop _all
15     qui set obs 500
16     qui gen time = _n
17     qui tsset time
18
19     qui gen double e = rnormal()
20     qui gen double y = e in 1
21     qui replace y = 0.4*L1.y + e + 0.5*L1.e in 2
22     qui replace y = 0.4*L1.y + 0.3*L2.y + e + 0.5*L1.e + 0.3*L2.e in 3/L
23     qui arima y, ar(1/2) ma(1)
24     post buffer (_b[ARMA:L1.ar]) (_b[ARMA:L2.ar]) (_b[ARMA:L1.ma])
25
26 }
27
28 postclose buffer
29 use arma22_misspecified.dta,clear
30
31 sum beta_L1 beta_L2 alpha_L1
32
33 set scheme white_tableau
34 graph set window fontface "Arial Narrow"
35
36 * AR(1) Plot
37 kdensity beta_L1, lcolor(navy) lpattern(solid) lwidth(medthick) ///
38     xline(0.4, lcolor(black) lpattern(shortdash)) ///
39     name(g1, replace) title("{bf:AR(1) Parameter}") ///
40     xtitle("Estimated Value") ///
41     xlabel(0 0.4 0.5 1 1.5 2 ,labsize(medium) nogrid) ///
42     ylabel(, labsize(medium) nogrid) note(" ")
43
44 * AR(2) Plot
45 kdensity beta_L2, lcolor(maroon) lpattern(solid) lwidth(medthick) ///
46     xline(0.3, lcolor(black) lpattern(shortdash)) ///
47     name(g2, replace) title("{bf:AR(2) Parameter}") ///
48     xtitle("Estimated Value") ///
49     xlabel(-1 -0.5 0 0.3 0.5 1 ,labsize(medium) nogrid) ///
50     ylabel(, labsize(medium) nogrid) note(" ")
51
52 * MA(1) Plot
53 kdensity alpha_L1, lcolor(forest_green) lpattern(solid) lwidth(medthick) ///
54     xline(0.5, lcolor(black) lpattern(shortdash)) ///
55     name(g3, replace) title("{bf:MA(1) Parameter}") ///
56     xtitle("Estimated Value") xlabel(,labsize(medium) nogrid) ///
57     note(" ") ylabel(, labsize(medium) nogrid)
58
59 * Combine Graph
60 graph combine g1 g2 g3, ///
61     rows(1) title("{bf:Simulation Results: ARMA(2,1)}") ///
62     xsize(8) ysize(4) ///
63     note("Black dashed line represents the true parameter") ///

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64     name(Biased, replace)
65
66 graph export Biased_arma.emf, name(Biased) replace
67
68
69
70 * (2) Unbiased Correct ARMA(2,2) Model Simulation
71 *-----
72
73 clear all
74 set seed 1000
75
76 capture postclose buffer
77 postfile buffer beta_L1 beta_L2 alpha_L1 alpha_L2 using arma22_correct_model.dta,
  replace
78
79 forvalues i = 1/1000 {
80     qui drop _all
81     qui set obs 500
82     qui gen time = _n
83     qui tsset time
84
85     qui gen double e = rnormal()
86     qui gen double y = e in 1
87     qui replace y = 0.4*L1.y + e + 0.5*L1.e in 2
88     qui replace y = 0.4*L1.y + 0.3*L2.y + e + 0.5*L1.e + 0.3*L2.e in 3/L
89     arima y, ar(1/2) ma(1/2)
90     matlist e(b)
91     post buffer (_b[ARMA:L.ar]) (_b[ARMA:L2.ar]) (_b[ARMA:L.ma]) (_b[ARMA:L2.ma])
92
93 }
94
95 postclose buffer
96 use arma22_correct_model.dta,clear
97
98 sum beta_L1 beta_L2 alpha_L1 alpha_L2
99
100 set scheme white_tableau
101 graph set window fontface "Arial Narrow"
102
103
104 * AR(1) Plot
105 kdensity beta_L1, lcolor(navy) lpattern(solid) lwidth(medthick) ///
106     xline(0.4, lcolor(black) lpattern(shortdash)) ///
107     name(g1, replace) title("{bf:AR(1) Parameter}") ///
108     xtitle("Estimated Value", size(medium)) ytitle(, size(medium)) note(" ") ///
109     xlabel(0 0.4 0.5 1 1.5, labsize(medium) nogrid) ylabel(, labsize(medium)
nogrid)
110
111 * AR(2) Plot
112 kdensity beta_L2, lcolor(maroon) lpattern(solid) lwidth(medthick) ///
113     xline(0.3, lcolor(black) lpattern(shortdash)) ///
114     name(g2, replace) title("{bf:AR(2) Parameter}") ///
115     xtitle("Estimated Value", size(medium)) ytitle(,size(medium)) note(" ") ///
116     xlabel(-0.5 0 0.3 0.5, labsize(medium) nogrid) ylabel(, labsize(medium)
nogrid)
117
118 * MA(1) Plot
119 kdensity alpha_L1, lcolor(forest_green) lpattern(solid) lwidth(medthick) ///
120     xline(0.5, lcolor(black) lpattern(shortdash)) ///
121     name(g3, replace) title("{bf:MA(1) Parameter}") ///
122     xtitle("Estimated Value", size(medium)) ytitle(,size(medium)) note(" ") ///
123     xlabel(-0.5 0 0.5 1, labsize(medium) nogrid) ylabel(, labsize(medium) nogrid)

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124
125 * MA(2) Plot
126 kdensity alpha_L2, lcolor(purple) lpattern(solid) lwidth(medthick) ///
127     xline(0.3, lcolor(black) lpattern(shortdash)) ///
128     name(g4, replace) title("{bf:MA(2) Parameter}") ///
129     xtitle("Estimated Value",size(medium)) ytitle(,size(medium)) note(" ") ///
130     xlabel(0 0.3 0.5, labsize(medium) nogrid) ylabel(, labsize(medium) nogrid)
131
132 * Combine into 2x2 Grid
133 graph combine g1 g2 g3 g4, ///
134     rows(2) cols(2) ///
135     title("{bf:Simulation Results: Correctly Specified ARMA(2,2)}") ///
136     note("Black dashed line represents the true parameter", size(small)) ///
137     xsize(8) ysize(4) name(CorrectModel, replace)
138
139
140 graph export ARMA_Correct.emf, name(CorrectModel) replace
141
142
143
```